IE 4461 Semester Project

The Effect of Energy Drinks on Heart Rate and Reaction Time

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Summary

The impact of different energy drinks (measured by caffeine content) will be evaluated by measuring the changes in heart rate and reaction time of participants before and after consumption. This experiment will evaluate both the physiological and cognitive effects of caffeine on college students, who are common consumers of energy drinks. By taking measurements of an individual's performance before and after consuming an energy drink, the strength of correlation between the variables can be determined. A range of caffeine values, from 80 mg to 300 mg (common drink contents include 160 mg and 200 mg) will be used to determine if the increase in caffeine has a particular relationship with heart rate and reaction time. Results from the experiment reveal that our hypotheses for both heart rate and reaction time were true. The more caffeine consumed through energy drinks, the more participants' heart rates increased, and reaction times lowered. Upon drinking 300 mg (the maximum amount from the experiment) of caffeine, heart rate increased by a mean of 6 beats per minute and reaction time decreased by an average of 34 milliseconds.

Background

In a college environment, all students are familiar with caffeine consumption, whether they themselves are consumers or know of consumers. Endless nights of studying, busy schedules, and extracurricular activities such as sports activities are all part of the college experience. To keep up, students tend to need assistance from tangible sources. The National Library of Medicine reported that 92% of students consume caffeine, with the overall population's average caffeine consumption being 159 mg/day. Many students consistently rely on caffeine for several reasons: to stay awake, social aspects of consumption, improve concentration, increase physical energy, improve mood, and alleviate stress (Malinauskas et al., 2007). Due to the increasing popularity of energy drinks, the impact of caffeine on certain health aspects is worth studying.

Caffeine is an adenosine receptor antagonist, which reduces drowsiness and temporarily heightens alertness. Heart rate, an indicator of cardiac response, is usually utilized in studies examining the potential negative effects of energy drinks. Conversely, a study performed on college students at Winona State University proved that 80mg of caffeine has little to no impact

on heart rate (Ragsdle et al., 2010). Though 80 mg is a smaller portion of caffeine compared to the typical energy drink. The recommended maximum amount of caffeine consumption is 400 mg, so this project will investigate varying mg doses to determine if there is a correlation (Mayo Clinic Staff, 2025). Another study has shown that heart rate and heart rate variability are unaffected by caffeine consumption (Leimrot, et al., 2023). This would be evidence that furthers the work from Ragsdle, despite their study being limited to 80 mg of caffeine.

Caffeine, as well as having potential impacts on heart rate, is also seen to increase alertness and reaction time. Caffeine has been shown to enhance cognitive performance and increase what's known as positive effect in humans (Hachenberger et al., 2025). Positive effect was defined as feelings associated with contentedness, enthusiasm, and happiness in caffeine consumers. Caffeine has also been shown to decrease reaction times, with these changes persisting for three to four hours (Michael et al., 2008). High consumers, but not moderate consumers, in experiments have demonstrated significantly faster reaction times relative to placebos (Attwood et al., 2007). Overall, reaction times have been found to decrease in individuals who have used caffeine. Lastly, caffeine can increase alertness in those with restricted sleep, though to an insignificant degree (Bartrim et al., 2020).

Generally, caffeine has been found to have a significant effect on users' reaction times and alertness levels in past studies. With extensive prior research, there are still notable gaps in literature. Few studies have compared multiple caffeine doses above 80 mg using the same population or investigated both the physiological and cognitive metrics, such as heart rate and reaction time. This experiment will address these gaps by analyzing varying caffeine levels and their effect on college students' heart rate and reaction time.

Objective and Hypothesis

The relationship between different types of energy drinks with physiological and cognitive performance will be evaluated by measuring heart rate and reaction time, comparing them to baseline metrics.

Null and alternative hypotheses for heart rate:

$$H_0: \mu_w = \mu_{caff}$$

(There is no difference in mean heart rate before and after caffeine consumption)

$$H_1: \mu_w < \mu_{caff}$$

(Mean heart rate increases after caffeine consumption)

Null and alternative hypotheses for reaction time:

$$H_0: \mu_w = \mu_{caff}$$

(There is no difference in mean reaction time before and after caffeine consumption)

$$H_1: \mu_w > \mu_{caff}$$

(Mean reaction time decreases after caffeine consumption, showing improvements)

Methods and Procedure

The experiment will consist of individuals taking several measurements of their heart rate and reaction time at different caffeine doses. Before any caffeine consumption, five measurements of the heart rate will be taken (one per minute), and five measurements of reaction time will be taken (one after the other). This data, before caffeine consumption, will serve as the control variable, and further measures will be compared against it.

After consuming the entirety of an energy drink (the independent variable: 80mg, 160mg, 200mg, or 300mg), the individual will wait 30 minutes, then have measurements taken of their heart rate and reaction time (dependent variables). According to the National Institutes of Health, 99% of caffeine is absorbed within 45 minutes of ingestion and can stay in the bloodstream for up to 10 hours. The rest period is crucial, so participants are asked to consume one energy drink per day, instead of multiple in one day, for this experiment.

Methods

Equipment:

- Different caffeinated beverages:
    - 8.4 oz. Red Bull (80 mg)
    - 16 oz. Monster (160 mg)
    - 12 oz. Celsius (200 mg)
    - 16 oz. Reign (300 mg)

- Watch that measures HR and Respiration
- Laptop for the Reaction Test: Human Benchmark - Reaction Time Test

Procedure

The study begins with twenty undergraduate student participants, who each will record their baseline heart rate and reaction time measurements using the watch and the benchmark test. To measure resting heart rate in bpm, they will take five measurements, one per minute. For standard reaction time in milliseconds, they will take five measurements using the website. Next, students will drink the energy drink provided. After a 30-minute rest period after complete consumption of the drink, the students will re-test their heart rates and reaction times using the prior steps. This process is repeated and recorded for all the caffeine doses. At least 10 hours must pass before the subject repeats the experiment. This process repeats with each of the 20 participants to complete the experiment.

Variables

Independent Variables

The independent variable is the amount of caffeine (mg)

- 8.4 oz. Red Bull (80 mg)
- 16 oz. Monster (160 mg)
- 12 oz. Celsius (200 mg)
- 16 oz. Reign (300 mg)

The control variable is 0 mg of caffeine: water/no drink

Dependent Variables

- Heart rate (bpm)
- Reaction time (ms)

Constant Variables

- Reaction time test website
- Quiet, non-distracting environment for the reaction time test
- No physical activity during heart rate measurement

- General age (college students)
- Rest period (30 minutes)

Results

Participant	Caffine Dosage (mg)	Trial #	Reaction Time (ms)	Time (min)	Heart Rate (bpm)
	0	1		1	
	0	2		2	
	0	3		3	
	0	4		4	
	0	5		5	
		Mean	**#DIV/0!**	**Mean**	**#DIV/0!**
	80	1		1	
	80	2		2	
	80	3		3	
	80	4		4	
	80	5		5	
		Mean	**#DIV/0!**	**Mean**	**#DIV/0!**
	160	1		1	
	160	2		2	
	160	3		3	
	160	4		4	
	160	5		5	
		Mean	**#DIV/0!**	**Mean**	**#DIV/0!**
	200	1		1	
	200	2		2	
	200	3		3	
	200	4		4	
	200	5		5	
		Mean	**#DIV/0!**	**Mean**	**#DIV/0!**
	300	1		1	
	300	2		2	
	300	3		3	
	300	4		4	
	300	5		5	
		Mean	**#DIV/0!**	**Mean**	**#DIV/0!**

Table 1: Data Sheet for Each Participant

Caffeine (mg)	Reaction Time (ms)	Heart Rate (bpm)
0	317.49	72.12
80	307.995	72.58
160	300.84	76.545
200	280.33	76.85
300	283.2	78.07

Table 2: Reaction Times and Heart Rate Means

Caffeine (mg)	RT (Reaction Time)	HR (Heart Rate)
0	69.7	12.29
80	63.94	12.09
160	56.95	11.12
200	41.23	10.38
300	35.83	10.56

Table 3: Reaction Time and Heart Rate Standard Deviations

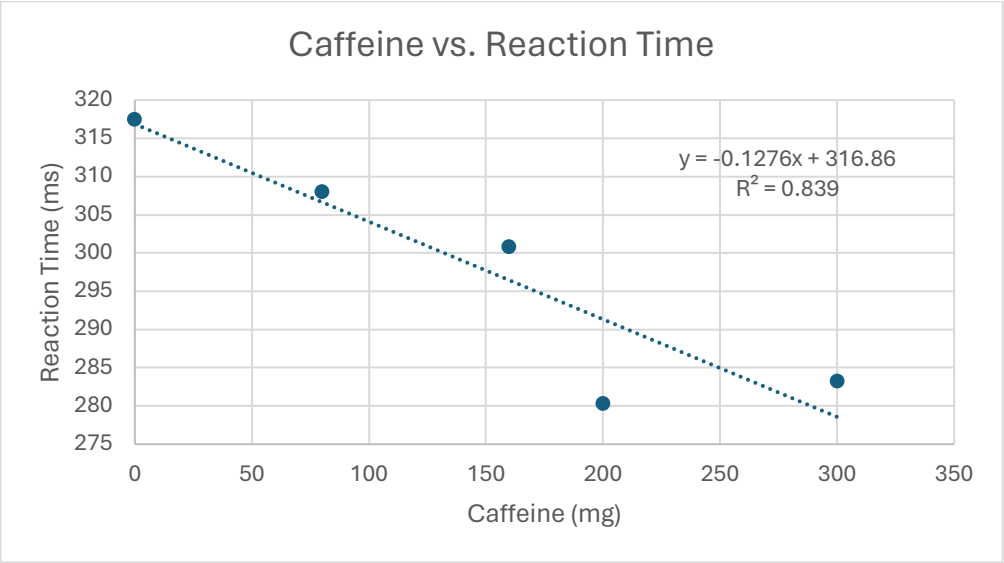


Figure 1: Caffeine vs. Reaction Time Line Graph

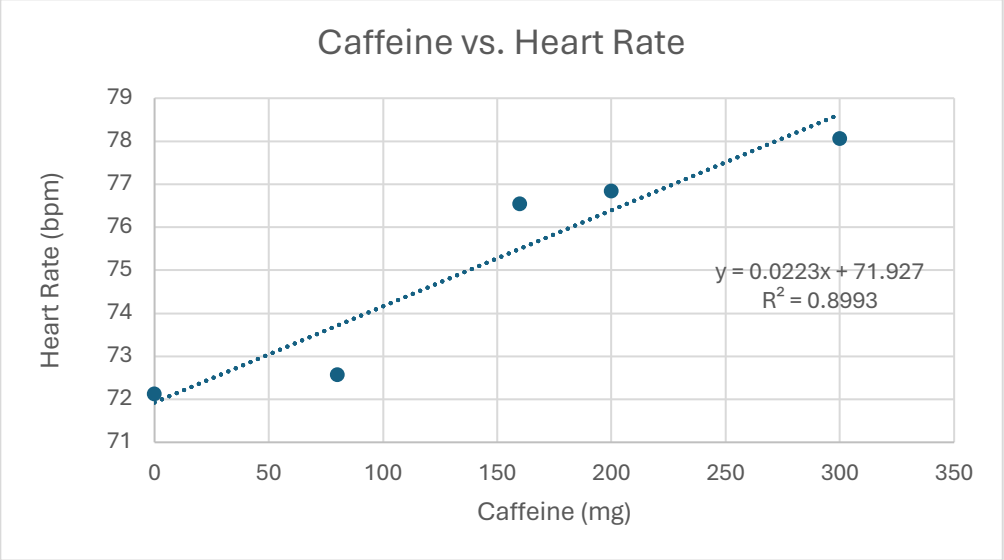


Figure 2: Caffeine vs. Heart Rate Line Graph

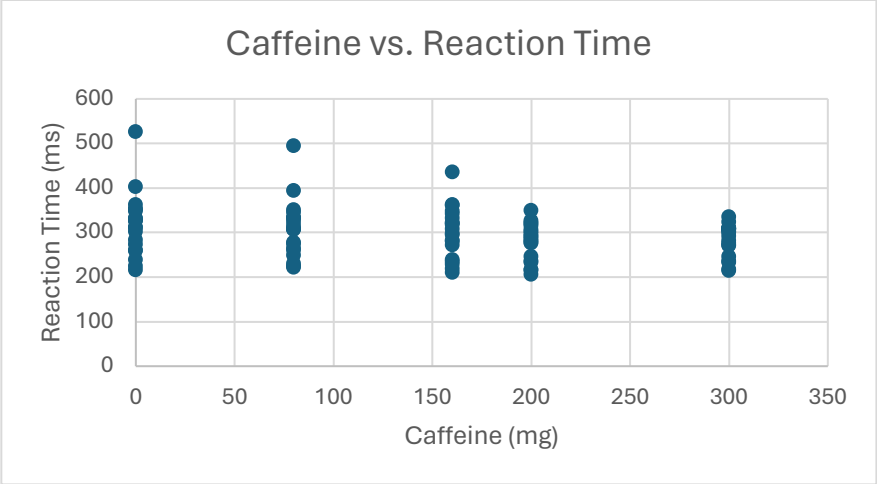


Figure 3: Caffeine vs. Reaction Time Scatter Plot

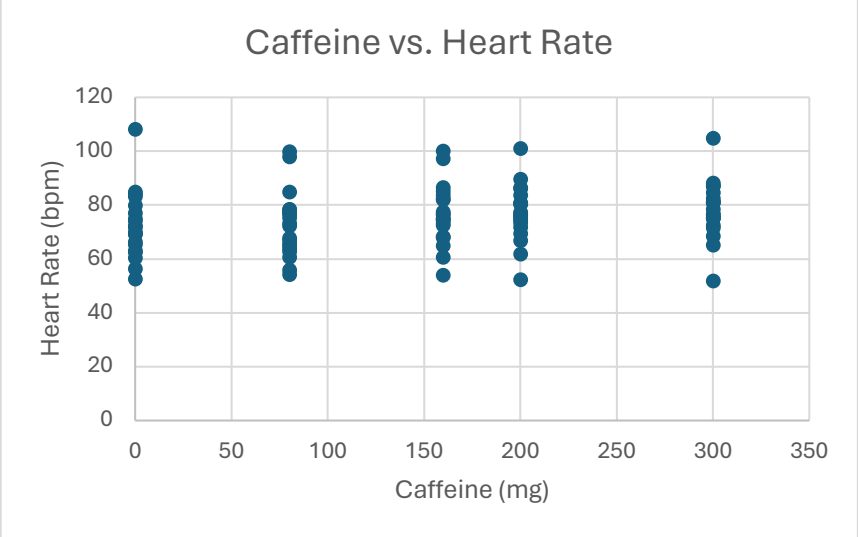


Figure 4: Caffeine vs. Heart Rate Scatter Plot

Discussion

Mean values for both reaction time and heart rate trended as expected. Reaction time started at 317 ms with the control caffeine amount (0mg) and ended at 283.2 ms at 300 mg of caffeine. Heart rate trended upwards, with mean values starting at 72 bpm with 0 mg of caffeine and ending at 78 bpm with 300 mg of caffeine. Compared with prior studies mentioned in the background section, the values obtained in this experiment remained true for reaction time means, but not for heart rate means.

The observed decrease in reaction time values due to increasing caffeine doses proves that energy drinks improve alertness. As stimulant levels increase due to caffeine consumption, cognitive processing speed increases as well. The findings align with caffeine's known role as an adenosine receptor antagonist (promoting wakefulness/reaction speed). Work by Attwood and Bartrim is supported by the findings of this experiment. Though reaction time decreased with higher doses of caffeine, the improvement rate decreased beyond 200 mg. This could imply a threshold where additional caffeine does not have a meaningful impact on humans.

The positive linear correlation between heart rate and caffeine doses reflects caffeine's known stimulatory effect on the brain. While the lab group's hypothesis and previous findings suggested modest increases, the steady rise from 72 bpm to 78 bpm reveals even mild caffeine doses significantly influence the brain's stimulatory activity. Unlike background studies conducted for reaction times, those done by both Ragsdle and Leimrot do not support the mean values found for heart rate. Ragsdle and Leimrot found that caffeine doses did not have a meaningful impact on heart rate, contrary to the findings. Perhaps this reflects energy drinks' specific effects on heart rate in opposition to sources such as coffee.

Standard deviation values for reaction time decreased from 70 ms at the control doses of 0 mg to 35.8 ms at 300 mg. This remarkable increase indicates that the participants' performance became more consistent at higher caffeine levels. These findings could imply caffeine enhances focus that may otherwise cause greater variation amongst the participants. Heart rate variability decreased to a lesser extent, though still suggesting a more uniform physiological response across those participating in the study. The caffeine's effect on the central nervous system could override some individual differences in sleep quality or baseline energy levels, molding a more standardized response profile with the experiment's subjects.

Conclusion

The results indicate that caffeine intake, especially from energy drinks, can lower reaction speed while increasing heart rate. Patterns in mean and standard deviation values imply that increasing doses of caffeine enhance alertness and readiness in a predictable way. Though the lowering improvements of reaction time beyond 200 mg could point to a threshold where additional caffeine yields little further benefit. At higher doses, variability decreases, revealing more stable

performance as caffeine doses increase. Further research could separate caffeine from other ingredients in the energy drink, which could lead to shaky results. It could also use larger samples to determine whether the effects the experiment found generalize beyond the specific energy drink tested.

Statistical Analysis

Tests to determine the statistical significance of the data have been performed via Excel. While the mean values of the data show a clear trend, further analysis was conducted to determine its validity. For both dependent variables, heart rate and reaction time, an output sheet shows the regression statistics, ANOVA table, and coefficients table. Results are shown in Figures 5 and 6, with explanations below each figure.

Regression Statistics								
Multiple R	0.202673261							
R Square	0.041076451							
Adjusted R Square	0.031291517							
Standard Error	11.16850107							
Observations	100							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	523.6303172	523.6303172	4.19792817	0.043146323			
Residual	98	12224.07078	124.7354162					
Total	99	12747.7011						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	71.92730945	1.96225518	36.65543105	5.08826E-59	68.03327791	75.821341	68.0332779	75.821341
X Variable 1	0.022335747	0.010901418	2.048884616	0.043146323	0.000702238	0.04396926	0.00070224	0.04396926

Figure 5: Excel Statistical Significance Analysis for Heart Rate

The Multiple R value is the correlation coefficient between caffeine and heart rate, and a value of 0.203 indicates a weak positive correlation. The R-squared value is 0.041, which means that about 4.1% of the variation in heart rate is explained by caffeine doses. The adjusted R-squared value accounts for the sample size but still shows a weak relationship. From the ANOVA table, p-value for this data is 0.043, which is less than the alpha value of 0.05, allowing the team to reject the null hypothesis. From this, the model is statistically significant, meaning that caffeine does have a measurable linear effect on heart rate. From the coefficients table, both p-values are less than 0.05, indicating that there is a statistically significant positive relationship (since the slope value is 0.022). As caffeine increases, heart rate increases, or higher caffeine

levels are associated with higher heart rates. Although the effect is statistically significant, the strength of the relationship is weak.

Regression Statistics								
Multiple R	0.235364968							
R Square	0.055396668							
Adjusted R Square	0.045757859							
Standard Error	54.546654							
Observations	100							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	17100.01453	17100.01453	5.747252117	0.018409535			
Residual	98	291583.0714	2975.337463					
Total	99	308683.0859						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	316.8616997	9.583600673	33.0629072	5.90325E-55	297.8433561	335.880043	297.843356	335.880043
X Variable 1	-0.127639863	0.053242226	-2.39734272	0.018409535	-0.233297327	-0.0219824	-0.2332973	-0.0219824

Figure 6: Excel Statistical Significance Analysis for Reaction Time

The Multiple R value is the correlation coefficient between caffeine and reaction time, and a value of 0.235 indicates a weak negative correlation. The R-squared value is 0.055, which means that about 5.5% of the variation in reaction time is explained by caffeine doses. The adjusted R-squared value accounts for the sample size but still shows a weak relationship. The p-value for this data is 0.018, which is less than the alpha value of 0.05, allowing the team to reject the null hypothesis. From the ANOVA table, the model is statistically significant, meaning that caffeine does have a measurable linear effect on reaction time. From the coefficients table, both p-values are less than 0.05, indicating that there is a statistically significant negative relationship (since the slope value is -0.12). As caffeine increases, reaction time decreases, and higher caffeine levels are associated with faster reactions. Despite the effect being statistically significant, the strength of the relationship is weak.

Multivariate

Correlations

	Caffeine	Reaction Time	Heart Rate
Caffeine	1.0000	-0.2354	0.2027
Reaction Time	-0.2354	1.0000	0.4053
Heart Rate	0.2027	0.4053	1.0000

Figure 7: JMP Correlation Analysis

The JMP output depicts a weak negative correlation between caffeine and reaction time (-0.2354) and a weak positive correlation (0.2027) between caffeine heart rate.

Other Considerations

When analyzing the results from the data, it is crucial to consider outside factors that may have influenced the results. For this project, the potential impacts of anxiety, ADHD, caffeine tolerance, consumption at specific times of day, emotional state, and biological aspects on performance were briefly researched for consideration.

A study was conducted to evaluate the relationship between caffeine intake and anxiety in healthy individuals. Caffeine acts as a central nervous system stimulant through adenosine receptor antagonism, increasing neuronal activity, norepinephrine release, and heart rate, which can heighten anxiety symptoms. When analyzed by doses, less than 400mg/day was associated with a moderate increase in anxiety. Caffeine can exacerbate anxiety effects in individuals with preexisting anxiety. (Liu, 2024).

In a study using animal models that exhibit ADHD symptoms, researchers examined the effects of caffeine on cognitive and cardiovascular behavior. The results of the study indicate that people with ADHD may experience fewer changes in their heart rate from caffeine than neurotypical individuals. It also indicates that individuals with ADHD may experience greater improvement in reaction time from caffeine than the general population. (Vázquez, 2022) While this experiment was not performed on similar subjects that the team is testing, it may have reasonable ties with participants' results who are diagnosed with ADHD.

A study conducted by individuals at the University of Edinburgh performed a double-blind crossover study, where adults consumed either 200 mg of caffeine or a placebo twice daily for a week before receiving a 200 mg caffeine challenge. Compared to the participants who did not know about caffeine, the regular users showed smaller increases in blood pressure and smaller decreases in blood flow. These results indicate that tolerance develops caffeine's cardiovascular and cerebral effects, but the responses are not eliminated. This means that consistent caffeine usage leads to partial tolerance, and people still experience measurable stimulant effects, but to a lesser effect. (Watson, 2002). This is relevant to the experiment because some participants indulge in energy drinks more often than others. Some participants consume an energy drink every day, while others wait until there's an exam or other circumstances that require a boost in

alertness. Regular consumers consume caffeine to restore normal functioning and relieve withdrawal symptoms.

Caffeine has the strongest impact on reaction time when consumed in the morning, typically when the individual is fatigued or sleep deprived. Individuals given caffeine at 7:00 AM and 7:00 PM had faster reaction times at 7:00 AM, indicating that caffeine helps offset morning sluggishness. Afternoon or evening caffeine intake will still improve reaction time, but adds less benefit, since natural alertness and reaction speed are already higher later in the day. (Zhang, 2024).

Caffeine impacts alertness, which can lead to feelings of happiness and energy at low doses. At higher doses, though, people can become more anxious, stressed, and experience other negative emotions that may affect their heart rate by increasing it. A reliable test found was inducing 5mg of caffeine per kg can recreate an arousal of mood (Herz, 1999)

The size of the heart is different depending on whether the participant is a man or a woman. Women have smaller hearts, leading to a lower cardiac output. Since the female heart pumps less blood out with each beat, it needs to beat faster to match the male's heart output. The normal heart rate is 70 bpm to 85 bpm. The average adult male heart rate is between 70 to 72 bpm, and for women it is 78 to 82 bpm. (Prabhavathi, 2014).

Limitations

The experiment has several limitations that potentially impacted the outcome of the results. One being the number of participants. Analyzing data of a randomly selected sample of 20 participants from an entire population only allows a broad conclusion. This limitation of participants could be problematic when we consider participants' differing caffeine tolerances. A participant's ability to process caffeine will affect their body's response and, therefore, their heart rate and reaction time measurements. As each individual repeats different trials of the experiment, they become more familiar with how to use the reaction time test. This could mean that their reaction time from their baseline measurements is higher than the rest, only because of a learning curve with using the website. Furthermore, the time of day when caffeine was consumed in the experiment varied across participants and trials, meaning caffeine affected everyone differently depending on their level of fatigue at the time of consumption. The limitations analyzed should be considered when interpreting the conclusion.

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